# 6.4 Uniformly Continuous Implies Continuous, Rule by Rule

Chapter 5 opened with a question. Continuity and uniform continuity differ only in where one quantifier stands; why does one imply the other in a few lines, while the converse is false? Section 5.1 gave the informal answer: a variable may depend on the variables quantified to its left, and only on them. Exercise (C1: Where the $\forall x$ stands) of Section 6.1 asked for both translations into formulas, and Section 6.2 negated uniform continuity (Example (Three negations from analysis) (c)). This section answers the question with the rules. It derives "uniformly continuous implies continuous" in full, line by line, with the bounded rules of Proposition 6.1.1. It then reads off the derivation the one thing this section adds: **which boxes are open where a constant is chosen**, and what that says about what the constant may depend on. The same reading locates the exact lines at which the converse breaks down, and the function $x^2$ shows that the converse is not merely unproved but false.

## 6.4.1 Two definitions that differ in one place

Let $A$ be a set of real numbers and $f$ a function defined on $A$. We work over the domain $\mathbb{R}$, with $f$ a one-place function symbol, $\in A$ a one-place predicate symbol (Section 5.2), and $|\cdot|$ the absolute value. Here the two definitions are only **quantifier patterns**, and nothing is used about $f$ except that $f(x)$ is a term. (A function symbol has a value at every real number; the values outside $A$ are never used, because $f$ is only applied to variables bounded by $\in A$, as with sequences in Section 5.1.)

$f$ is **uniformly continuous on** $A$ (UC) when

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \forall y \in A\ \big(|x - y| < \delta \to |f(x) - f(y)| < \varepsilon\big).$$

$f$ is **continuous at every point of** $A$ (C) when

$$\forall x \in A\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall y \in A\ \big(|x - y| < \delta \to |f(x) - f(y)| < \varepsilon\big).$$

The two formulas have the same matrix and the same four bounded quantifiers. Only $\forall x \in A$ has moved: in C it stands **before** $\exists \delta > 0$, in UC **after** it. In the game of Section 5.1, the defender chooses $\delta$ in C after seeing $x$ and $\varepsilon$, and in UC after seeing only $\varepsilon$. So in UC a single $\delta$ must serve every point of $A$ at once: that is what **uniformly** means.

*Quick check.* Swap the first two quantifiers of UC, to get $\exists \delta > 0\ \forall \varepsilon > 0\ \forall x \in A\ \forall y \in A\ (|x - y| < \delta \to |f(x) - f(y)| < \varepsilon)$. What does the new formula say about two points of $A$ closer than $\delta$? Does $f(x) = x$ on $\mathbb{R}$ satisfy it?

*Solution.* Now $\delta$ is chosen before $\varepsilon$, so it must work for every $\varepsilon > 0$ at once. For two points $x, y \in A$ with $|x - y| < \delta$, this says $|f(x) - f(y)| < \varepsilon$ for every $\varepsilon > 0$, and by Proposition 6.2.5, $f(x) = f(y)$. So the formula says: for some $\delta > 0$, any two points of $A$ closer than $\delta$ have the **same** value. The function $f(x) = x$ on $\mathbb{R}$ fails it: whatever $\delta > 0$ is proposed, the points 0 and $\delta/2$ are closer than $\delta$ and have different values. Yet $f(x) = x$ is uniformly continuous, with $\delta = \varepsilon$. Swapping $\forall \varepsilon$ and $\exists \delta$ asks far too much.

## 6.4.2 The skeleton first

Hold $\varepsilon$ fixed, and write $K(x,\delta)$ for "$\delta$ works at $x$", the formula $\forall y \in A\ (|x-y| < \delta \to |f(x)-f(y)| < \varepsilon)$. Drop the bounds on $\delta$ and $x$ for a moment. Then UC at this $\varepsilon$ says $\exists\delta\,\forall x\,K(x,\delta)$, one $\delta$ for every point, and C at this $\varepsilon$ says $\forall x\,\exists\delta\,K(x,\delta)$, a $\delta$ for each point. The step from the first to the second is Proposition 5.6.2 with other letters:

| | | | | |
|---|---|---|---|---|
| 1 | $\exists\delta\,\forall x\,K(x,\delta)$ | | | |
| 2 | $x_0$ | $\delta_0$ | $\forall x\,K(x,\delta_0)$ | asm, for $\exists$E |
| 3 | | | $K(x_0,\delta_0)$ | $\forall$E 2 |
| 4 | | | $\exists\delta\,K(x_0,\delta)$ | $\exists$I 3 |
| 5 | | $\exists\delta\,K(x_0,\delta)$ | | $\exists$E 1, 2–4 |
| 6 | $\forall x\,\exists\delta\,K(x,\delta)$ | | | $\forall$I 2–5 |

Fix a point, obtain the one $\delta$ that works everywhere, choose it for this point. The converse, from $\forall x\,\exists\delta\,K$ to $\exists\delta\,\forall x\,K$, is the swap whose search stalled in Section 5.6. The rest of this section puts $\varepsilon$ and the bounds back and writes $K$ out. That costs five lines: the premise UC itself (line 1), the assumptions $x_0 \in A$, $\varepsilon_0 > 0$ and $\delta_0 > 0$ (lines 2, 3 and 5), and the line that closes the box of $\varepsilon_0$ (line 10). The skeleton's line 1 becomes line 4 of the derivation below (UC at $\varepsilon_0$), its lines 2–5 become lines 6–9, and its line 6 becomes line 11, which closes the box of $x_0$.

*Quick check.* Why may line 5 of the skeleton close the box of $\delta_0$?

*Solution.* $\delta_0$ does not appear in line 5, $\exists\delta\,K(x_0,\delta)$, nor in the existential of line 1, which is also the only premise, and the one box still open, that of $x_0$, has no assumption. Every clause of $\exists$E holds.

## 6.4.3 The derivation

**Example. Uniformly continuous implies continuous**

Derive C from UC, with the bounded rules of Proposition 6.1.1.

*Idea.* Run the search method of Section 6.3. The goal C begins $\forall x \in A$, then $\forall\varepsilon > 0$: the **goal-directed** steps open two bounded $\forall$I boxes, "let $x_0 \in A$" and "let $\varepsilon_0 > 0$", and write the last two lines. The new goal begins $\exists\delta > 0$, and no witness is in sight, so it is **postponed** (Section 5.5). Turn to the premise UC. It begins $\forall\varepsilon > 0$, and the term the goal mentions is $\varepsilon_0$, whose bound $\varepsilon_0 > 0$ is on a line: $\forall$E at $\varepsilon_0$ (Section 5.3). That gives an $\exists\delta > 0$ line, and its $\exists$E box is opened **at once**, with a witness $\delta_0$ (Section 5.6), aiming for the postponed goal as its last line. Inside, the premise has become "$\delta_0$ works for all $x, y \in A$"; instantiate it at the point we have, $x_0$. Now the postponed goal has its witness: $\delta_0$ itself.

*Before reading on: which lines does the line introducing $\delta_0$ rest on, and does any of them mention $x_0$?*

*Solution.*

| | | | |
|---|---|---|---|
| 1 | | $\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ \forall y \in A\ (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon)$ | |
| 2 | $x_0$ | $x_0 \in A$ | asm, for $\forall$I |
| 3 | $\varepsilon_0$ | $\varepsilon_0 > 0$ | asm, for $\forall$I |
| 4 | | $\exists\delta > 0\ \forall x \in A\ \forall y \in A\ (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon_0)$ | $\forall$E 1, 3 |
| 5 | $\delta_0$ | $\delta_0 > 0$ | asm, for $\exists$E |
| 6 | | $\forall x \in A\ \forall y \in A\ (\lvert x - y\rvert < \delta_0 \to \lvert f(x) - f(y)\rvert < \varepsilon_0)$ | asm, for $\exists$E |
| 7 | | $\forall y \in A\ (\lvert x_0 - y\rvert < \delta_0 \to \lvert f(x_0) - f(y)\rvert < \varepsilon_0)$ | $\forall$E 6, 2 |
| 8 | | $\exists\delta > 0\ \forall y \in A\ (\lvert x_0 - y\rvert < \delta \to \lvert f(x_0) - f(y)\rvert < \varepsilon_0)$ | $\exists$I 5, 7 |
| 9 | | $\exists\delta > 0\ \forall y \in A\ (\lvert x_0 - y\rvert < \delta \to \lvert f(x_0) - f(y)\rvert < \varepsilon_0)$ | $\exists$E 4, 5–8 |
| 10 | | $\forall\varepsilon > 0\ \exists\delta > 0\ \forall y \in A\ (\lvert x_0 - y\rvert < \delta \to \lvert f(x_0) - f(y)\rvert < \varepsilon)$ | $\forall$I 3–9 |
| 11 | | $\forall x \in A\ \forall\varepsilon > 0\ \exists\delta > 0\ \forall y \in A\ (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon)$ | $\forall$I 2–10 |

Every line is tagged with its rule; lines 4, 7 and 8 are the one-line bounded rules, and lines 9, 10 and 11 close bounded boxes. The terms and the side conditions:

- Line 4: $\forall$E at $\varepsilon_0$, with its bound on line 3. The term $\varepsilon_0$ is a variable that does not appear in line 1, so it is free for $\varepsilon$.
- Line 7: $\forall$E at $x_0$, with its bound on line 2, which is available because the box of line 2 is still open.
- Line 8: $\exists$I with the witness $\delta_0$, its bound on line 5 and the instance on line 7.
- Line 9 ($\exists$E, flag $\delta_0$): $\delta_0$ does not appear in line 9 (the witness was forgotten at line 8), in line 4, in the premise, or in the assumptions still open at line 9, lines 2 and 3.
- Line 10 ($\forall$I, flag $\varepsilon_0$): $\varepsilon_0$ does not appear in the premise, in line 10, or in the assumption still open at line 10, line 2.
- Line 11 ($\forall$I, flag $x_0$): $x_0$ does not appear in the premise or in line 11, and no assumption is open.

The answer to the prompt: line 5 is the assumption of the $\exists$E box, so it rests on line 4, which rests on lines 1 and 3. None of the three mentions $x_0$. The witness $\delta_0$ was obtained from the premise at $\varepsilon_0$ alone, before $x_0$ played any part. That is uniformity, read off the lines.

The derivation has 11 lines. Written with the unbounded rules it has 17, since each of its six bounded steps (lines 4, 7, 8, 9, 10 and 11) costs one line more (Section 6.1). The bounded rules keep it at the level at which a mathematician thinks, and the same argument in prose is three sentences, each aligned with the lines it abbreviates:

> Let $x \in A$ and $\varepsilon > 0$ [lines 2, 3, 10, 11]. By uniform continuity there is $\delta > 0$ such that $\lvert u - v\rvert < \delta$ implies $\lvert f(u) - f(v)\rvert < \varepsilon$ for all $u, v \in A$; choose such a $\delta$ [4, 5, 6, 9]. Applied with $u = x$, this gives $\lvert f(x) - f(y)\rvert < \varepsilon$ for every $y \in A$ with $\lvert x - y\rvert < \delta$, so this $\delta$ works for $x$ and $\varepsilon$ [7, 8].

The prose renames the bound variables of UC to $u, v$, so that "$x$" can name the fixed point; a derivation needs no such care, because $\forall$E handles bound variables by substitution.

## 6.4.4 Reading dependence off the boxes

The derivation shows something the informal proof leaves unsaid: the exact place where $\delta$ is chosen, and what surrounds it there.

*A constant chosen in a box may depend on the names of the boxes open around that box, and it does depend only on the names in the line it comes from.*

*Quick check.* In the derivation above, which flagged boxes surround the box of $\delta_0$, opened at line 5? What may $\delta_0$ therefore depend on, and what does it in fact depend on?

*Solution.* The boxes of $x_0$ (opened at line 2) and of $\varepsilon_0$ (opened at line 3). So $\delta_0$ may depend on $x_0$ and on $\varepsilon_0$, as C allows. In fact it depends only on $\varepsilon_0$: it comes from line 4, which mentions $\varepsilon_0$ and not $x_0$.

Why "may"? Inside the boxes of $x_0$ and $\varepsilon_0$, every term built from $x_0$ and $\varepsilon_0$ is available, so a premise could have been instantiated at such a term, and a witness obtained from the resulting $\exists$ line would depend on both names. The side conditions permit that here, because the goal C puts $\exists\delta$ inside both $\forall$'s: the witness is forgotten by $\exists$I (line 8) before either box closes. Why "does"? Because everything known about $\delta_0$ is what its $\exists$ line says, and line 4 mentions $\varepsilon_0$ alone.

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↻ **Second reading** · *Choosing the delta before the point*

So $\delta_0$ could have been obtained before $x_0$ entered at all: fix $\varepsilon_0$, obtain $\delta_0$, and only then let $x_0 \in A$. That is the order of UC itself. A derivation in that order nests the boxes as $\varepsilon_0$, $\delta_0$, $x_0$ (the box of $\delta_0$ rests on line 4, so it stays inside the box of $\varepsilon_0$), and it ends in $\forall\varepsilon > 0\ \forall x \in A\ \exists\delta > 0\ \forall y \in A\ (\dots)$, which is C with its first two quantifiers exchanged. C follows by exchanging them back, as in the swap of Example (Flagged boxes at work) (c), here with the bounded rules: boxes for $x_1 \in A$ and $\varepsilon_1 > 0$, $\forall$E at $\varepsilon_1$ and then at $x_1$, and two $\forall$I lines. C's order of choices allows more dependence than this proof uses.

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The limit derivation of Section 6.1 showed the same thing on a smaller scale: its witness $N_0$ came from a line obtained at $\varepsilon_0/2$, inside the box of $\varepsilon_0$ only, and so it depended on $\varepsilon_0$.

## 6.4.5 The converse breaks at a side condition

The converse, from C to UC, is the interesting direction. The search for it fails, and the failure has an exact address.

**Example. The search from C to UC**

Run the search method on the sequent with premise C and goal UC, and find where it stops.

*Try it first (ten minutes): start the search. Where does a $\delta$ come from, and at what moment can you get one?*

*Idea.* The obvious plan copies the derivation above: fix $\varepsilon_0$, obtain $\delta_0$ from the premise, choose it as the witness. It runs into an obstacle at the second step. In C, $\exists\delta$ stands behind $\forall x \in A$, so to reach it the premise must first be instantiated at a point, and the bounded $\forall$E needs a line $t \in A$. No such line exists until we assume one, in a box for some $x_0$. The $\delta_0$ obtained inside that box comes from a line that mentions $x_0$, and both side conditions then hold it in: it cannot leave its own box while a line about it still mentions it, and $x_0$ cannot be generalized while a line about $\delta_0$ still mentions $x_0$.

*Solution.* The goal UC begins $\forall\varepsilon > 0$, so open a box for $\varepsilon_0$. The new goal $\exists\delta > 0\ \forall x \in A\ \forall y \in A\ (\dots)$ is postponed. The premise C begins $\forall x \in A$, and no line $t \in A$ is available; the only way to get one is to assume it, so open a bounded box "let $x_0 \in A$". Then the premise yields a $\delta$. Here is how far the search gets, every line legal:

| | | |
|---|---|---|
| 1 | $\forall x \in A\ \forall\varepsilon > 0\ \exists\delta > 0\ \forall y \in A\ (\lvert x-y\rvert < \delta \to \lvert f(x)-f(y)\rvert < \varepsilon)$ | |
| 2 | $\varepsilon_0$ \| $\varepsilon_0 > 0$ | asm, for $\forall$I |
| 3 | $x_0$ \| $x_0 \in A$ | asm, for $\forall$I |
| 4 | $\forall\varepsilon > 0\ \exists\delta > 0\ \forall y \in A\ (\lvert x_0-y\rvert < \delta \to \lvert f(x_0)-f(y)\rvert < \varepsilon)$ | $\forall$E 1, 3 |
| 5 | $\exists\delta > 0\ \forall y \in A\ (\lvert x_0-y\rvert < \delta \to \lvert f(x_0)-f(y)\rvert < \varepsilon_0)$ | $\forall$E 4, 2 |
| 6 | $\delta_0$ \| $\delta_0 > 0$ | asm, for $\exists$E |
| 7 | $\forall y \in A\ (\lvert x_0-y\rvert < \delta_0 \to \lvert f(x_0)-f(y)\rvert < \varepsilon_0)$ | asm, for $\exists$E |
| 8 | $\exists\delta > 0\ \forall y \in A\ (\lvert x_0-y\rvert < \delta \to \lvert f(x_0)-f(y)\rvert < \varepsilon_0)$ | $\exists$I 6, 7 |
| 9 | $\exists\delta > 0\ \forall y \in A\ (\lvert x_0-y\rvert < \delta \to \lvert f(x_0)-f(y)\rvert < \varepsilon_0)$ | $\exists$E 5, 6–8 |
| 10 | $\forall x \in A\ \exists\delta > 0\ \forall y \in A\ (\lvert x-y\rvert < \delta \to \lvert f(x)-f(y)\rvert < \varepsilon_0)$ | $\forall$I 3–9 |
| 11 | $\forall\varepsilon > 0\ \forall x \in A\ \exists\delta > 0\ \forall y \in A\ (\lvert x-y\rvert < \delta \to \lvert f(x)-f(y)\rvert < \varepsilon)$ | $\forall$I 2–10 |

The goal inside the box of $\varepsilon_0$ needs a $\delta$ that works for **all** points: the line $\forall x \in A\ \forall y \in A\ (\lvert x-y\rvert < \delta_0 \to \lvert f(x)-f(y)\rvert < \varepsilon_0)$, so that $\exists$I at $\delta_0$ could finish. Line 7 says this only for the point $x_0$. There are two ways to try to generalize it, and each is stopped by a side condition.

- **Generalize inside the box of $\delta_0$.** After line 7, open a new flagged box with the name $x_0$ and the assumption $x_0 \in A$, repeat line 7 in it, and close it by $\forall$I with the conclusion $\forall x \in A\ \forall y \in A\ (\lvert x-y\rvert < \delta_0 \to \dots)$. That line breaks the side condition of $\forall$I: $x_0$ appears in assumptions still open there, line 3 and line 7. (With a different name $x_1$ the box is legal, but line 7 says nothing about $x_1$, so the box cannot reach its last line.)
- **Leave the box of $\delta_0$ first, carrying line 7.** Close the $\exists$E box with line 7 itself as its conclusion, in order to generalize over $x_0$ outside it. That line breaks the side condition of $\exists$E: its conclusion mentions the witness $\delta_0$.

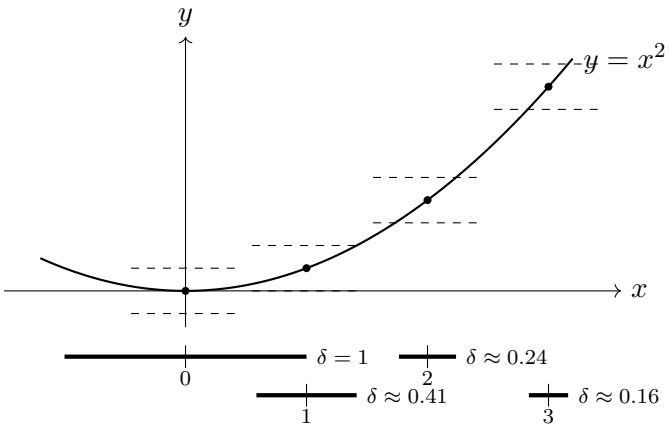
The only legal way out of the box of $\delta_0$ is to forget $\delta_0$, as line 8 does with $\exists$I before line 9 closes the box. Then $\forall$I on $x_0$ is legal, and line 10 is what it gives: $\forall x \in A\ \exists\delta > 0$, with $\delta$ still behind $x$. That is C at $\varepsilon_0$ again, not UC, and closing the box of $\varepsilon_0$ gives line 11, which is C with its first two quantifiers exchanged. The search has gone round in a circle.

These are the same two blocks that stopped the converse of Proposition 5.6.2: a ∀I blocked by an open assumption that mentions its name, and an ∃E box that cannot close on a line mentioning its witness. That is no coincidence: with the bounds and the matrix stripped away, C to UC is the swap $\forall x\,\exists\delta$ to $\exists\delta\,\forall x$.

A failed search proves nothing about all derivations; a counter-reading does, and the next subsection gives one: a function for which C is true and UC false. Section 7.2 explains why that settles the question.

## 6.4.6 A function that is continuous and not uniformly continuous

Here is the picture first; the example proves it.




With $\varepsilon = 1$, the largest $\delta$ that works at $x \geq 0$ is $\sqrt{x^2+1} - x$: 1 at 0, about 0.41 at 1, 0.24 at 2, 0.16 at 3. It shrinks as $x$ grows, and no single positive $\delta$ works at every $x$.

**Example. The square function on the real line**

Let $f(x) = x^2$ on $A = \mathbb{R}$. Show that $f$ satisfies C and does not satisfy UC.

*Solution. C holds.* Let $x \in \mathbb{R}$ and $\varepsilon > 0$. Let

$$\delta := \min\Big(1,\ \frac{\varepsilon}{2|x|+1}\Big),$$

which is positive, since $\varepsilon > 0$ and $2|x| + 1 \geq 1 > 0$. It depends on $\varepsilon$ **and on** $x$. Let $y \in \mathbb{R}$ with $|x - y| < \delta$. By the triangle inequality (Example (The triangle inequality by cases)), used twice, and since $\delta \leq 1$, $|y| \leq |x| + |y - x| < |x| + 1$, so $|x + y| \leq |x| + |y| < 2|x| + 1$. Hence

$$|x^2 - y^2| = |x - y| \cdot |x + y| \leq |x - y|\,(2|x| + 1) < \delta\,(2|x| + 1) \leq \varepsilon,$$

where the strict step uses $2|x| + 1 > 0$, and the last step uses $\delta \leq \varepsilon/(2|x| + 1)$. So this $\delta$ works for $x$ and $\varepsilon$.

*UC fails.* By the negation of uniform continuity in Example (Three negations from analysis)

(c), with the bounds $x \in \mathbb{R}$ and $y \in \mathbb{R}$ dropped because they always hold, we must show

$$\exists \varepsilon > 0 \; \forall \delta > 0 \; \exists x \; \exists y \; (|x - y| < \delta \wedge |x^2 - y^2| \geq \varepsilon).$$

Take $\varepsilon = 1$. Let $\delta > 0$ be arbitrary, and take $x := 1/\delta$ and $y := 1/\delta + \delta/2$, which depend on $\delta$. Then $|x - y| = \delta/2 < \delta$, and, since $y > x > 0$ (so $y - x$ and $y + x$ are both positive),

$$|x^2 - y^2| = (y - x)(y + x) = \frac{\delta}{2}\Big(\frac{2}{\delta} + \frac{\delta}{2}\Big) = 1 + \frac{\delta^2}{4} \geq 1.$$

Since $\delta$ was arbitrary, UC fails.

The $\delta$ for C grows smaller as $|x|$ grows, and no single $\delta$ serves the whole line. The proof of failure says why: however small $\delta$ is, far enough out (at $x = 1/\delta$) the graph is so steep that two points $\delta/2$ apart have values more than 1 apart. Read in the real numbers, with $f$ as squaring, the premise C is true and the conclusion UC is false: a counter-reading, and by Section 7.2 no derivation of UC from C exists.

*Warning.* **"Depending on $\varepsilon$" is a claim, and it can be false.** When a proof says "choose $\delta$ (depending on $\varepsilon$)", the parenthesis records which boxes are open where $\delta$ is chosen, and which names its formula uses. This is the logical origin of the book's rule that every chosen constant says what it depends on: the dependence is a record of the open boxes, and a reader can check it. A proof of uniform continuity must be able to write "depending on $\varepsilon$" truthfully. A "proof" that $x^2$ is uniformly continuous on $\mathbb{R}$ that takes $\delta = \min(1, \varepsilon/(2|x|+1))$ "(depending on $\varepsilon$)" has a formula for $\delta$ that mentions $x$: the $\delta$ was chosen inside the box of $x$, so the proof shows only C, and Example (The square function on the real line) shows that UC is false.

The proof that UC fails for $x^2$ established, with numbers, the positive form of the negation from Section 6.2. That form, taken before its last order fact, refutes UC by logic alone, and a derivation shows how; one TAUT line (Section 6.3) carries its propositional core.

**Example. The negation at work, in one derivation**

Let $E$ be $\exists \varepsilon > 0 \; \forall \delta > 0 \; \exists x \in A \; \exists y \in A \; (|x - y| < \delta \wedge \neg(|f(x) - f(y)| < \varepsilon))$, the form of Example (Three negations from analysis) (c) before its last order fact. Derive $E \vdash \neg$UC, with the bounded rules of Section 6.1.

*Solution.*

| | | | |
|---|---|---|---|
| 1 | | $\exists\varepsilon > 0\ \forall\delta > 0\ \exists x \in A\ \exists y \in A\ (\lvert x-y\rvert < \delta \wedge \neg(\lvert f(x)-f(y)\rvert < \varepsilon))$ | |
| 2 | | $\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ \forall y \in A\ (\lvert x-y\rvert < \delta \to \lvert f(x)-f(y)\rvert < \varepsilon)$ | asm, for ¬I |
| 3 | $\varepsilon_1$ | $\varepsilon_1 > 0$ | asm, for ∃E |
| 4 | | $\forall\delta > 0\ \exists x \in A\ \exists y \in A\ (\lvert x-y\rvert < \delta \wedge \neg(\lvert f(x)-f(y)\rvert < \varepsilon_1))$ | asm, for ∃E |
| 5 | | $\exists\delta > 0\ \forall x \in A\ \forall y \in A\ (\lvert x-y\rvert < \delta \to \lvert f(x)-f(y)\rvert < \varepsilon_1)$ | ∀E 2, 3 |
| 6 | $\delta_1$ | $\delta_1 > 0$ | asm, for ∃E |
| 7 | | $\forall x \in A\ \forall y \in A\ (\lvert x-y\rvert < \delta_1 \to \lvert f(x)-f(y)\rvert < \varepsilon_1)$ | asm, for ∃E |
| 8 | | $\exists x \in A\ \exists y \in A\ (\lvert x-y\rvert < \delta_1 \wedge \neg(\lvert f(x)-f(y)\rvert < \varepsilon_1))$ | ∀E 4, 6 |
| 9 | $x_1$ | $x_1 \in A$ | asm, for ∃E |
| 10 | | $\exists y \in A\ (\lvert x_1-y\rvert < \delta_1 \wedge \neg(\lvert f(x_1)-f(y)\rvert < \varepsilon_1))$ | asm, for ∃E |
| 11 | $y_1$ | $y_1 \in A$ | asm, for ∃E |
| 12 | | $\lvert x_1-y_1\rvert < \delta_1 \wedge \neg(\lvert f(x_1)-f(y_1)\rvert < \varepsilon_1)$ | asm, for ∃E |
| 13 | | $\forall y \in A\ (\lvert x_1-y\rvert < \delta_1 \to \lvert f(x_1)-f(y)\rvert < \varepsilon_1)$ | ∀E 7, 9 |
| 14 | | $\lvert x_1-y_1\rvert < \delta_1 \to \lvert f(x_1)-f(y_1)\rvert < \varepsilon_1$ | ∀E 13, 11 |
| 15 | | $\bot$ | TAUT 12, 14 |
| 16 | | $\bot$ | ∃E 10, 11–15 |
| 17 | | $\bot$ | ∃E 8, 9–16 |
| 18 | | $\bot$ | ∃E 5, 6–17 |
| 19 | | $\bot$ | ∃E 1, 3–18 |
| 20 | | $\neg\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ \forall y \in A\ (\lvert x-y\rvert < \delta \to \lvert f(x)-f(y)\rvert < \varepsilon)$ | ¬I 2–19 |

Line 15's propositional skeleton is $p \wedge \neg q,\ p \to q \models \bot$, so the line replaces ∧E twice, →E and ¬E, four lines. The boxes on the witness lines 5, 8 and 10 are opened as soon as those lines exist, the move "name the witness early" (Section 5.6), which Section 6.3 puts first. Every closing line's witness ($y_1$, $x_1$, $\delta_1$, $\varepsilon_1$) is absent from $\bot$, from the ∃ line it cites, from the premise, and from the assumptions still open.

## 6.4.7 Witness functions

UC says that for every $\varepsilon > 0$ some $\delta > 0$ works for all points. A rule that picks such a $\delta$ for each $\varepsilon$ is a function, $\delta(\varepsilon)$: a **witness function** for UC. For $f(x) = x/2$ on $\mathbb{R}$, $\delta(\varepsilon) = 2\varepsilon$ works, since $\lvert x/2 - y/2\rvert = \lvert x-y\rvert/2 < \varepsilon$ when $\lvert x-y\rvert < 2\varepsilon$.

For C the witness may depend on the point too: $\delta(x, \varepsilon)$. For $x^2$, Example (The square function on the real line) gave $\delta(x, \varepsilon) = \min(1, \varepsilon/(2\lvert x\rvert + 1))$.

A derivation shows which arguments a witness function needs. A witness obtained inside the boxes of $\varepsilon_0$ and $x_0$ may be a function of both, and one obtained inside the box of $\varepsilon_0$ alone is a function of $\varepsilon_0$ alone. "May depend" and "does depend" are exactly the two readings of "Reading dependence off the boxes". So UC holds when some witness function for C ignores its

point argument, and the derivation of Example (Uniformly continuous implies continuous) does the reverse: it turns a $\delta(\varepsilon)$ for UC into a $\delta(x, \varepsilon) \coloneqq \delta(\varepsilon)$ for C, one that ignores $x$.

In the game of Section 5.1, a witness function is a strategy for the defender: a rule that answers every challenge, and the defender wins by following it.

A formula of this book cannot say "there is a function": its variables range over numbers. A function symbol can stand for one chosen function, and that is how Sections 5.1 and 6.1 wrote sequences.

A tempting "proof" of the converse tries to build a witness function for UC out of one for C, by taking the smallest of its values.

**Example. A "proof" that continuity gives uniform continuity**

Find the error in this argument, which claims that every function continuous at every point of $A$ is uniformly continuous on $A$.

> *Let $\varepsilon > 0$. For each $x \in A$, continuity at $x$ gives $\delta_x > 0$ with $|f(x) - f(y)| < \varepsilon$ whenever $y \in A$ and $|x - y| < \delta_x$. Let $\delta$ be the smallest of the numbers $\delta_x$. Then $\delta > 0$, and it works at every point, so $f$ is uniformly continuous.*

*Solution.*

- The conclusion is false for $x^2$ on $\mathbb{R}$ (Example (The square function on the real line)), so some step fails, and it is the third sentence.
- Infinitely many positive numbers need not have a smallest, and they may come arbitrarily close to 0. With $\varepsilon = 1$, the largest $\delta$ that works for $x^2$ at $x \geq 0$ is $\sqrt{x^2+1} - x$ (the figure above), which is $1/(\sqrt{x^2+1} + x) < 1/(2x)$ for $x > 0$, and so smaller than any given positive number once $x$ is large.
- In the rules: each $\delta_x$ is the witness of an $\exists$E box inside the box of $x_0$, and "the smallest of them" would be a line about all the witnesses at once. Using it needs a $\delta$ that works at every point, the line $\forall x \in A\, \forall y \in A\, (|x - y| < \delta \to \dots)$ whose two routes Example (The search from C to UC) found blocked by side conditions.
- For finite $A = \{a_1, \dots, a_k\}$ the proof is right: $k$ uses of $\exists$E give $\delta_1, \dots, \delta_k$ (the subsection "Two witnesses" of Section 5.6, repeated), and $\min(\delta_1, \dots, \delta_k) > 0$ works at every point.

The search stalls at the same side condition that stopped Example (The search from C to UC); that $x^2$ is a counter-reading for the converse is the evidence that no other route exists (Section 7.2).

*Quick check.* For which sets $A$ does the step "let $\delta$ be the smallest of the $\delta_x$" make sense, and give a positive $\delta$?

*Solution.* For finite non-empty $A$. Finitely many positive numbers have a smallest, and it is positive. For empty $A$, UC holds outright, since its $\forall x \in A$ is vacuous.

*Remark* (Looking ahead). Chapter 8 shows how the eleven lines of Example (Uniformly continuous implies continuous) become the three sentences of a written proof, at several levels of detail.

## 6.4.8 Exercises

### 6.4.8.1 A. Check your understanding

**Exercise. A1**

These parts are about the derivation of Example (Uniformly continuous implies continuous).

(a) Why is line 7 legal? Name the rule, the term, and the lines it cites, and say why those lines are available.

(b) At which line is the premise's "for all $x \in A$" used, and at which term?

(c) State the idea of the derivation in two sentences.

(d) Run the derivation for $f(x) = 3x + 1$ on $A = \mathbb{R}$, where UC holds with $\delta = \varepsilon/3$. For a given $x_0$ and $\varepsilon_0 > 0$, which $\delta$ does the derivation hand to C at line 8, and does it depend on $x_0$? Check that it works.

*Solution.* *(a)* Line 7 is the bounded ∀E at the term $x_0$. It cites line 6, $\forall x \in A\, \forall y \in A\, (\dots)$, and line 2, $x_0 \in A$, the bound's instance. Line 6 is the assumption of the ∃E box that contains line 7, and line 2 is the assumption of the box of $x_0$, which also contains line 7; both boxes are still open at line 7. The term $x_0$ is a variable not appearing in line 6, so it is free for $x$.

*(b)* At line 7, at the term $x_0$. Line 6 says that $\delta_0$ works for **all** pairs of points of $A$, and line 7 uses that for pairs with first point $x_0$.

*(c)* Uniform continuity at $\varepsilon$ supplies one $\delta$ that works for all points at once, and it is supplied before any point is named. Continuity at a point $x$ then takes that same $\delta$, specialized to the point $x$.

*(d)* Line 4 says that some $\delta > 0$ works at $\varepsilon_0$, and line 8 hands C that witness $\delta_0$, whatever it is; the derivation knows only lines 5 and 6 about it. Here $\varepsilon_0/3$ is such a witness (so is any $\delta$ with $0 < \delta \leq \varepsilon_0/3$). Reading the derivation with the witness the exercise names, $\delta_0 = \varepsilon_0/3$, the $\delta$ handed to C is $\varepsilon_0/3$. It does not depend on $x_0$. It works: for every real $y$ with $|x_0 - y| < \varepsilon_0/3$,

$$|f(x_0) - f(y)| = |(3x_0 + 1) - (3y + 1)| = 3|x_0 - y| < 3 \cdot \frac{\varepsilon_0}{3} = \varepsilon_0.$$

**Exercise. A2: Drill**

Each part takes a line or two. The parts run from single steps, through a missing line, a justification column and a step to judge, to witness functions, a negation and one short derivation. UC and C are the formulas of "Two definitions that differ in one place".

(a) Line 1 is UC and line 2 is $1 > 0$. Write the line that the bounded ∀E 1, 2 gives.

(b) The available lines are $\delta_1 > 0$ and $\forall y \in A\ (|x_0 - y| < \delta_1 \to |f(x_0) - f(y)| < \varepsilon_0)$. Write the line that the bounded $\exists$I with witness $\delta_1$ gives.

(c) In Example (The negation at work, in one derivation), line 13 is the bounded $\forall$E from line 7 at $x_1$. Which second line must be available?

(d) In the six-line derivation of "The skeleton first", give the rule and the cited lines of lines 3 and 4.

(e) In Example (The search from C to UC), give the rule and the cited lines of line 10.

(f) Is this correct? Inside the boxes of $\varepsilon_0$ and $x_0$, a witness $\delta_0$ obtained from C at $x_0$ is used in $\exists$I to write $\exists \delta > 0\ \forall x \in A\ \forall y \in A\ (\dots)$.

(g) Give a witness function for UC of $f(x) = 5x$ on $\mathbb{R}$.

(h) For $x^2$, what does the witness function $\min(1, \varepsilon/(2|x| + 1))$ give at $x = 3$, $\varepsilon = 1$?

(i) For $x^2$ and $\varepsilon = 1$, what is the largest $\delta$ that works at $x = 4$, by the formula under the figure?

(j) Negate C, "continuous at every point of $A$", into positive form.

(k) Which quantifier of C must move, and past which, to turn C into UC?

(l) Derive $\forall \varepsilon > 0\ \exists \delta > 0\ (\delta \le \varepsilon)$ with the bounded rules, given the arith line $\varepsilon_0 \le \varepsilon_0$.

*Solution.*

(a) $\exists \delta > 0\ \forall x \in A\ \forall y \in A\ (|x - y| < \delta \to |f(x) - f(y)| < 1)$.

(b) $\exists \delta > 0\ \forall y \in A\ (|x_0 - y| < \delta \to |f(x_0) - f(y)| < \varepsilon_0)$.

(c) Line 9, $x_1 \in A$.

(d) Line 3: $\forall$E 2, at $x_0$. Line 4: $\exists$I 3, with witness $\delta_0$.

(e) The bounded $\forall$I, 3–9.

(f) No. That $\exists$I needs the line $\forall x \in A\ \forall y \in A\ (|x - y| < \delta_0 \to \dots)$, and that line needs $\forall$I over $x_0$ while an open assumption about $\delta_0$ mentions $x_0$: the side condition of $\forall$I.

(g) $\delta(\varepsilon) = \varepsilon/5$, since $|5x - 5y| = 5|x - y| < \varepsilon$ when $|x - y| < \varepsilon/5$.

(h) $\min(1, 1/7) = 1/7$.

(i) $\sqrt{17} - 4 \approx 0.123$.

(j) $\exists x \in A\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists y \in A\ (|x - y| < \delta \wedge |f(x) - f(y)| \ge \varepsilon)$.

(k) $\forall x \in A$ moves to the right, past $\exists \delta > 0$. Passing $\forall \varepsilon > 0$ is harmless (two $\forall$'s exchange); passing $\exists \delta$ is the swap that fails.

(l) The witness function is $\delta(\varepsilon) = \varepsilon$:

| | | | |
|---|---|---|---|
| 1 | $\varepsilon_0$ \| $\varepsilon_0 > 0$ | | asm, for $\forall$I |
| 2 | \| $\varepsilon_0 \le \varepsilon_0$ | | arith |
| 3 | \| $\exists \delta > 0\ (\delta \le \varepsilon_0)$ | | $\exists$I 1, 2 |
| 4 | $\forall \varepsilon > 0\ \exists \delta > 0\ (\delta \le \varepsilon)$ | | $\forall$I 1–3 |

### 6.4.8.2 B. Practice

**Exercise. B1: Lipschitz implies uniformly continuous**

A function $f$ is **Lipschitz** on $A$ when

$$\exists L \geq 0\ \forall x \in A\ \forall y \in A\ (|f(x) - f(y)| \leq L \cdot |x - y|).$$

Prove that a function that is Lipschitz on $A$ is uniformly continuous on $A$, first as a derivation with the bounded rules (lines of arithmetic justified "arith"), then as a short proof in prose. Say what your $\delta$ depends on. (The bound $L \geq 0$ is part of the definition: if $A$ has at most one point, then $x = y$ in the matrix, and any $L$, even $L = -1$, would satisfy it. The bound rules this out, so a proof may use $L \geq 0$, for instance to divide by $L + 1$.)

*Solution.* The goal UC begins $\forall \varepsilon > 0$: open a box for $\varepsilon_0$. The new goal $\exists \delta > 0 \ldots$ is postponed until a witness is in hand. The premise begins $\exists L \geq 0$: open its $\exists$E box at once, with a witness $L_0$, aiming for the postponed goal, which does not mention $L_0$. Work backwards from what the boxes for $x_0$ and $y_0$ will need: from $|x_0 - y_0| < \delta$ we want $|f(x_0) - f(y_0)| < \varepsilon_0$, and the premise gives $|f(x_0) - f(y_0)| \leq L_0|x_0 - y_0| < L_0\delta$ when $L_0 > 0$. So $\delta = \varepsilon_0/L_0$ is the natural guess, but $L_0 = 0$ is allowed and then $\varepsilon_0/L_0$ is undefined. Take $\delta = \varepsilon_0/(L_0 + 1)$ instead: $L_0 + 1 \geq 1 > 0$, and $L_0\delta = \varepsilon_0 \cdot L_0/(L_0 + 1) < \varepsilon_0$.

| | | | |
|---|---|---|---|
| 1 | | $\exists L \geq 0\ \forall x \in A\ \forall y \in A\ (\lvert f(x) - f(y)\rvert \leq L \cdot \lvert x - y\rvert)$ | |
| 2 | $\varepsilon_0$ | $\varepsilon_0 > 0$ | asm, for $\forall$I |
| 3 | | $L_0$ $L_0 \geq 0$ | asm, for $\exists$E |
| 4 | | $\forall x \in A\ \forall y \in A\ (\lvert f(x) - f(y)\rvert \leq L_0 \cdot \lvert x - y\rvert)$ | asm, for $\exists$E |
| 5 | | $\varepsilon_0/(L_0{+}1) > 0$ | arith 2, 3 |
| 6 | | $x_0$ $x_0 \in A$ | asm, for $\forall$I |
| 7 | | $y_0$ $y_0 \in A$ | asm, for $\forall$I |
| 8 | | $\lvert x_0 - y_0\rvert < \varepsilon_0/(L_0{+}1)$ | asm, for $\to$I |
| 9 | | $\forall y \in A\ (\lvert f(x_0) - f(y)\rvert \leq L_0 \cdot \lvert x_0 - y\rvert)$ | $\forall$E 4, 6 |
| 10 | | $\lvert f(x_0) - f(y_0)\rvert \leq L_0 \cdot \lvert x_0 - y_0\rvert$ | $\forall$E 9, 7 |
| 11 | | $\lvert f(x_0) - f(y_0)\rvert < \varepsilon_0$ | arith 2, 3, 8, 10 |
| 12 | | $\lvert x_0 - y_0\rvert < \varepsilon_0/(L_0{+}1) \to \lvert f(x_0) - f(y_0)\rvert < \varepsilon_0$ | $\to$I 8–11 |
| 13 | | $\forall y \in A\ (\lvert x_0 - y\rvert < \varepsilon_0/(L_0{+}1) \to \lvert f(x_0) - f(y)\rvert < \varepsilon_0)$ | $\forall$I 7–12 |
| 14 | | $\forall x \in A\ \forall y \in A\ (\lvert x - y\rvert < \varepsilon_0/(L_0{+}1) \to \lvert f(x) - f(y)\rvert < \varepsilon_0)$ | $\forall$I 6–13 |
| 15 | | $\exists \delta > 0\ \forall x \in A\ \forall y \in A\ (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon_0)$ | $\exists$I 5, 14 |
| 16 | | $\exists \delta > 0\ \forall x \in A\ \forall y \in A\ (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon_0)$ | $\exists$E 1, 3–15 |
| 17 | | $\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \forall y \in A\ (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon)$ | $\forall$I 2–16 |

Line 11 is arithmetic from lines 2, 3, 8 and 10: $|f(x_0) - f(y_0)| \leq L_0|x_0 - y_0| \leq L_0 \cdot \varepsilon_0/(L_0{+}1)$, where the second step uses $L_0 \geq 0$, and $L_0/(L_0{+}1) < 1$ with $\varepsilon_0 > 0$ gives $L_0\varepsilon_0/(L_0{+}1) < \varepsilon_0$. Line 15 is the bounded $\exists$I at the term $\varepsilon_0/(L_0 + 1)$, with its bound on line 5. At line 16, $L_0$

does not appear in line 16 (the witness term was forgotten at line 15), in line 1, which is the premise, or in the assumption $\varepsilon_0 > 0$ still open there. At lines 13, 14 and 17 the flags $y_0$, $x_0$, $\varepsilon_0$ do not appear in the premise, in the line itself, or in the assumptions then open (among lines 2, 3, 4 and 6).

In prose: *Let $L \geq 0$ be a constant with $|f(x) - f(y)| \leq L|x - y|$ for all $x, y \in A$. Let $\varepsilon > 0$, and let $\delta := \varepsilon/(L+1)$, which is positive and depends on $\varepsilon$ (and on the constant $L$, which is fixed before $\varepsilon$), but not on any point. For $x, y \in A$ with $|x-y| < \delta$, $|f(x)-f(y)| \leq L|x-y| \leq L\delta = \varepsilon L/(L+1) < \varepsilon$. So $f$ is uniformly continuous on $A$.* The box of $L_0$ is open where $\delta$ is chosen, so $\delta$ may depend on $L_0$. That is harmless: $L_0$ is a constant of the function, not a point, and its box could as well have been opened first, outside the box of $\varepsilon_0$.

**Exercise. B2: Put the proof in order**

Below are ten sentences. Eight of them, in a suitable order, form a proof that a function $f$ uniformly continuous on $A$ is continuous at every point of $A$; two are distractors. Write down an order for the eight, and say why each distractor is wrong.

(i) Since $x$ and $y$ lie in $A$ and $|x - y| < \delta$, the choice of $\delta$, applied to $u = x$ and $v = y$, gives $|f(x) - f(y)| < \varepsilon$.

(ii) Let $\delta > 0$ be arbitrary.

(iii) Let $\varepsilon > 0$.

(iv) Since $\varepsilon > 0$ and $x \in A$ were arbitrary, $f$ is continuous at every point of $A$.

(v) Suppose $f$ is uniformly continuous on $A$.

(vi) By uniform continuity, applied to $\varepsilon$, choose $\delta > 0$ such that for all $u, v \in A$, $|u - v| < \delta$ implies $|f(u) - f(v)| < \varepsilon$.

(vii) Choose $\delta > 0$ for this $x$ and this $\varepsilon$, and then let $x \in A$ be arbitrary.

(viii) Let $x \in A$.

(ix) So for every $y \in A$, $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon$: this $\delta$ works for $x$ and $\varepsilon$.

(x) Let $y \in A$ with $|x - y| < \delta$.

*Solution.* An admissible order: (v), (viii), (iii), (vi), (x), (i), (ix), (iv).

(v) is the hypothesis of the implication being proved, so it opens the proof (its $\to$I box is the outermost one). After it, other orders are admissible as long as each sentence comes after what it uses. (vi) uses the $\varepsilon$ of (iii); (x) uses the $\delta$ of (vi) and the $x$ of (viii); then (i), (ix), (iv) follow in that order. So exactly three orders are admissible: (viii), (iii), (vi); or (iii), (viii), (vi); or (iii), (vi), (viii), each preceded by (v) and followed by (x), (i), (ix), (iv). Only the first matches the derivation of Example (Uniformly continuous implies continuous) directly. The second and the third fix $\varepsilon$ before $x$; the third even chooses $\delta$ before any point is given, which is possible because uniform continuity supplies $\delta$ from $\varepsilon$ alone. They correspond to variants of that derivation in which the box of $\varepsilon_0$ (in the third, also that of $\delta_0$) is opened outside the box of $x_0$. Both variants end in $\forall \varepsilon > 0\ \forall x \in A\ \exists \delta > 0\ \forall y \in A\ (\dots)$, and C follows by exchanging the two bounded universal quantifiers (the swap of Example (Flagged boxes at work) (c), with the bounded rules); sentence (iv), "since $\varepsilon$ and $x$ were arbitrary", covers both steps in prose.

Distractor (ii) introduces $\delta$ as an arbitrary object, by $\forall$I. But in "continuous at $x$" the $\delta$ is under $\exists$: the proof must **produce** it, by a choice from the hypothesis, not receive it from a challenger. A proof that took $\delta$ arbitrary would have to show that every $\delta > 0$ works, which is false in general.

Distractor (vii) chooses $\delta$ and only afterwards lets $x$ be arbitrary. It looks legitimate, because when (vii) is read after (viii), $x$ is already fixed, and continuity at $x$ does allow $\delta$ to depend on $x$. But the sentence reverses the order of the two steps: a $\delta$ fitted to one particular $x$ is followed by "let $x$ be arbitrary", which claims that the same $\delta$ works for every $x$. In the rules: if $\delta$ really was chosen for this $x$, its $\exists$E box assumes a line that mentions $x$, and generalizing over $x$ while that assumption is open breaks the side condition of $\forall$I, the first blocked line of Example (The search from C to UC). If $\delta$ does not depend on $x$, as here, then "for this $x$" is idle and the sentence should be (vi).

**Exercise. B3: The square root**

Prove that $f(x) = \sqrt{x}$ is uniformly continuous on $A = [0, \infty)$. You may use the school inequality $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$ for all $x, y \ge 0$. State what your $\delta$ depends on.

*Solution.* Let $\varepsilon > 0$, and let $\delta := \varepsilon^2$, which is positive and depends on $\varepsilon$ only. Let $x, y \in [0, \infty)$ with $|x - y| < \delta$. Then, by the inequality given and because the square root is strictly increasing on $[0, \infty)$,

$$|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|} < \sqrt{\varepsilon^2} = \varepsilon.$$

Since $x$ and $y$ were arbitrary, this $\delta$ works for all pairs of points; since $\varepsilon$ was arbitrary, $f$ is uniformly continuous on $[0, \infty)$. The $\delta$ was chosen before the points: in a derivation of this proof, its $\exists$I line would lie inside the box of $\varepsilon_0$ only.

**Exercise. B4: Witness functions**

For each function, give a witness function for uniform continuity on the set named, say what it depends on, and prove that it works.

(a) $f(x) = |x|$ on $\mathbb{R}$.
(b) $f(x) = x^2$ on $[0, 3]$.
(c) Say in one sentence why the formula of (b) gives no witness function on all of $\mathbb{R}$.

*Solution.* *(a)* $\delta(\varepsilon) = \varepsilon$, which depends on $\varepsilon$ only. Let $x, y \in \mathbb{R}$ with $|x - y| < \varepsilon$. By the triangle inequality (Example (The triangle inequality by cases)), $|x| \le |x - y| + |y|$ and $|y| \le |y - x| + |x|$, so $|x| - |y| \le |x - y|$ and $|y| - |x| \le |x - y|$; hence $||x| - |y|| \le |x - y| < \varepsilon$.

*(b)* $\delta(\varepsilon) = \varepsilon/6$. For $x, y \in [0, 3]$, $|x + y| = x + y \le 6$, so if $|x - y| < \varepsilon/6$, then

$$|x^2 - y^2| = |x - y| \cdot |x + y| \le 6|x - y| < \varepsilon.$$

It depends on $\varepsilon$ and on the bound 3 of the set, and on no point.

*(c)* On $\mathbb{R}$, $|x + y|$ has no bound, so no constant can replace the 6, and Example (The square

function on the real line) shows that no witness function exists there at all.

**Marking scheme.** (10 points) - 3: (a), with $||x| - |y|| \leq |x - y|$ proved (for instance by the triangle inequality twice). - 4: (b): 1 for the bound $|x + y| \leq 6$, 2 for the chain, 1 for the dependence. - 3: (c). Fatal error (0 for the step): a "witness" that mentions $x$ or $y$.

#### 6.4.8.3 C. Going deeper

**Exercise. C1: Composition, half done**

Let $f$ be uniformly continuous on $A$, with $f(x) \in B$ for every $x \in A$, and let $g$ be uniformly continuous on $B$. Here is the start of a proof that the composite $x \mapsto g(f(x))$ is uniformly continuous on $A$:

> Fix $\varepsilon > 0$. By the uniform continuity of $g$, choose $\eta > 0$ such that $|u - v| < \eta$ implies $|g(u) - g(v)| < \varepsilon$ for all $u, v \in B$.

The rest is blank. *Name the move, then complete it.*

*Solution* (Hint 1). The goal asks for a $\delta$. Which hypothesis can produce one, and at which positive number should it be applied?

*Solution* (Hint 2). "By the uniform continuity of $f$, applied to $\eta$, choose $\delta > 0$ such that ..."

*Solution. The move* is fix, obtain, choose: $\varepsilon$ was fixed ($\forall$I), $\eta$ obtained from the hypothesis on $g$ ($\forall$E at $\varepsilon$, then $\exists$E); now obtain $\delta$ from the hypothesis on $f$ at the term the goal needs, $\eta$ ($\forall$E at $\eta$, then $\exists$E), and choose that $\delta$ as the witness ($\exists$I).

*Completion.* Since $\eta > 0$, the uniform continuity of $f$, applied to $\eta$, gives $\delta > 0$ such that $|x-y| < \delta$ implies $|f(x)-f(y)| < \eta$ for all $x, y \in A$; choose such a $\delta$. It depends on $\eta$, hence on $\varepsilon$, and on no point. Let $x, y \in A$ with $|x-y| < \delta$. Then $|f(x)-f(y)| < \eta$, and $f(x), f(y) \in B$, so the choice of $\eta$, applied to $u = f(x)$ and $v = f(y)$, gives $|g(f(x)) - g(f(y))| < \varepsilon$. Since $x, y$ were arbitrary, this $\delta$ works for all pairs of points of $A$; since $\varepsilon > 0$ was arbitrary, $x \mapsto g(f(x))$ is uniformly continuous on $A$.

**Marking scheme.** (10 points) - 2: names the move (fix, obtain, choose), or the rules $\forall$E at $\eta$, $\exists$E, $\exists$I. - 3: applies the uniform continuity of $f$ at $\eta$ (not at $\varepsilon$), using $\eta > 0$, and chooses $\delta$ before any point. - 3: for arbitrary $x, y \in A$ with $|x - y| < \delta$, gets $|f(x) - f(y)| < \eta$ and then $|g(f(x)) - g(f(y))| < \varepsilon$, using $f(x), f(y) \in B$. - 2: concludes with the quantifiers in the order $\varepsilon$, $\delta$, points, and says $\delta$ depends on $\varepsilon$ only. Fatal errors (0 for the step): $\delta$ chosen after $x$ or depending on $x$; the uniform continuity of $g$ applied to points not known to lie in $B$.

**Exercise. C2: Gluing two sets**

Let $A$ and $B$ be sets of real numbers and $S$ the set of numbers that lie in $A$ or in $B$. Suppose $f$ is uniformly continuous on $A$ and uniformly continuous on $B$.

(a) Show that $f$ need not be uniformly continuous on $S$: take $A = (0,1)$, $B = (1,2)$, and $f(x) = 0$ for $x < 1$, $f(x) = 1$ for $x \geq 1$.

(b) Prove that $f$ is uniformly continuous on $S$ when $A = [a,b]$ and $B = [b,c]$ with $a < b < c$.

*Solution* (Hint 1). For (b), the two sets share the point $b$. Two points on opposite sides of $b$ are each close to $b$.

*Solution* (Hint 2). For (b), apply the two hypotheses at $\varepsilon/2$ and take $\delta = \min(\delta_1, \delta_2)$.

*Solution.* *(a)* On $A = (0,1)$, $f$ is constantly 0, so every $\delta > 0$ works for every $\varepsilon > 0$: $|f(x) - f(y)| = 0 < \varepsilon$. The same holds on $B = (1,2)$, where $f$ is constantly 1. On $S$ we prove the negation of uniform continuity (Example (Three negations from analysis) (c)). Take $\varepsilon = 1$. Let $\delta > 0$, and let $\eta := \min(\delta/4, \frac{1}{2})$, $x := 1 - \eta$, $y := 1 + \eta$. Then $\frac{1}{2} \leq x < 1$, so $x \in A \subseteq S$; $1 < y \leq \frac{3}{2}$, so $y \in B \subseteq S$; $|x - y| = 2\eta \leq \delta/2 < \delta$; and $|f(x) - f(y)| = |0 - 1| = 1 \geq \varepsilon$. So $f$ is not uniformly continuous on $S$. The gluing fails because the two sets come arbitrarily close at 1 but have no point in common: points close to 1 on opposite sides are each controlled only by their own set, and no common point links them.

*(b)* Let $\varepsilon > 0$. By the uniform continuity on $A$ at $\varepsilon/2$, choose $\delta_1 > 0$ such that $|f(x) - f(y)| < \varepsilon/2$ for all $x, y \in A$ with $|x - y| < \delta_1$; likewise choose $\delta_2 > 0$ for $B$. Let $\delta := \min(\delta_1, \delta_2) > 0$, which depends on $\varepsilon$ only. Let $x, y \in S$ with $|x - y| < \delta$. There are three cases.

- *Case 1:* $x, y \in A$. Then $|f(x) - f(y)| < \varepsilon/2 < \varepsilon$, since $|x - y| < \delta \leq \delta_1$.
- *Case 2:* $x, y \in B$. The same, with $\delta_2$.
- *Case 3: one point in each set*, say $x \in A$ and $y \in B$ (the other way round is the same with the letters exchanged). Then $x \leq b \leq y$, so $|x - b| = b - x \leq y - x = |x - y| < \delta \leq \delta_1$, and $b \in A$; hence $|f(x) - f(b)| < \varepsilon/2$. Likewise $|b - y| \leq |x - y| < \delta_2$ and $b \in B$, so $|f(b) - f(y)| < \varepsilon/2$. By the triangle inequality (Example (The triangle inequality by cases)),
$$|f(x) - f(y)| \leq |f(x) - f(b)| + |f(b) - f(y)| < \varepsilon.$$

Every point of $S$ lies in $A$ or in $B$, so the cases cover all pairs. So $f$ is uniformly continuous on $S$. The common point $b$ is what (a) lacked: there, $A$ and $B$ came arbitrarily close at 1 but had no point in common.

**Marking scheme.** (10 points) - 3: (a) uniform continuity on $A$ and on $B$ checked, and the negation on $S$ proved: $\varepsilon$ first, then for arbitrary $\delta$ two points depending on $\delta$, with all three conditions checked. - 2: (b) $\delta_1$ for $A$ at some $\varepsilon_1 > 0$ and $\delta_2$ for $B$ at some $\varepsilon_2 > 0$ with $\varepsilon_1 + \varepsilon_2 \leq \varepsilon$ (the solution takes $\varepsilon_1 = \varepsilon_2 = \varepsilon/2$), and $\delta = \min(\delta_1, \delta_2)$, depending on $\varepsilon$ only. - 2: the cases with both points in one set. - 3: the mixed case, through the point $b$, with $|x - b|, |b - y| \leq |x - y|$ justified and the triangle inequality cited. Fatal errors (0 for the step): a $\delta$ depending on $x$ or $y$; the mixed case omitted; in (a), points chosen before $\delta$.

**Exercise. C3: Exit check: a quantifier derivation**

(a) Give a full derivation of

$$\exists x\,(P(x) \wedge \forall y\,R(x,y)) \vdash \forall y\,\exists x\,(P(x) \wedge R(x,y)),$$

and check the side condition of every line that closes a flagged box.

(b) For the converse, $\forall y\,\exists x\,(P(x) \wedge R(x,y)) \vdash \exists x\,(P(x) \wedge \forall y\,R(x,y))$, give a domain with two elements and a reading of $P$ and $R$ under which the premise is true and the conclusion false, checking both with the informal reading of Section 5.1.

*Solution. (a)* The goal is a $\forall$: open a flagged box for $y_0$. The premise is an $\exists$: open its $\exists$E box at once, with a witness $x_0$, aiming for the goal's instance. Inside, take the conjunction apart, instantiate $\forall y$ at $y_0$, and choose $x_0$ as the witness:

| | | | | |
|---|---|---|---|---|
| 1 | | | $\exists x\,(P(x) \wedge \forall y\,R(x,y))$ | |
| 2 | $y_0$ | $x_0$ | $P(x_0) \wedge \forall y\,R(x_0,y)$ | asm, for $\exists$E |
| 3 | | | $P(x_0)$ | $\wedge$E 2 |
| 4 | | | $\forall y\,R(x_0,y)$ | $\wedge$E 2 |
| 5 | | | $R(x_0,y_0)$ | $\forall$E 4 |
| 6 | | | $P(x_0) \wedge R(x_0,y_0)$ | $\wedge$I 3, 5 |
| 7 | | | $\exists x\,(P(x) \wedge R(x,y_0))$ | $\exists$I 6 |
| 8 | | | $\exists x\,(P(x) \wedge R(x,y_0))$ | $\exists$E 1, 2–7 |
| 9 | | | $\forall y\,\exists x\,(P(x) \wedge R(x,y))$ | $\forall$I 2–8 |

- Line 8 ($\exists$E, flag $x_0$): $x_0$ does not appear in line 8, in line 1 (the premise), or in any assumption still open at line 8 (there is none: the box of $y_0$ has no assumption). Line 7, the last line of the box of $x_0$, is a line of that box itself and is the formula $\exists x\,(P(x) \wedge R(x,y_0))$ that line 8 repeats (Definition 5.6.1).
- Line 9 ($\forall$I, flag $y_0$): $y_0$ does not appear in the premise or in line 9, and no assumption is open. Line 8, the last line of the box of $y_0$, is the instance at $y_0$ of $\exists x\,(P(x) \wedge R(x,y))$.

The other nesting, with the box of $y_0$ inside the box of $x_0$, works too: $\forall$I at the end of the inner box (its name $y_0$ is in no open assumption, since the assumption on line 2 does not mention it), then $\exists$E on a conclusion without $x_0$.

*(b)* Take a domain with two elements $a$, $b$; let $P$ be true of both, and let $R(x,y)$ mean "$x$ is $y$". The premise is true: for $y = a$ the witness $x = a$ has $P(a)$ and $R(a,a)$, and for $y = b$ the witness $x = b$ works. The conclusion is false: $x = a$ fails $\forall y\,R(a,y)$ at $y = b$, and $x = b$ fails it at $y = a$. The witness in the premise depends on $y$; the conclusion asks for one $x$ that serves every $y$. (Section 7.2 makes this informal reading a structure, and shows that it proves the converse is not derivable.)

**Marking scheme.** (10 points) - 2: a flagged box with a fresh name $y_0$ for the $\forall$I, closed correctly. - 3: an $\exists$E box with a fresh witness $x_0$, whose conclusion does not contain $x_0$, with the side condition checked. - 2: $\wedge$E, $\forall$E at $y_0$ and $\exists$I with the witness $x_0$, with the correct terms. - 3: the two-element reading: domain, meaning of $P$ and $R$, the premise checked for each $y$ and the conclusion refuted for each $x$. Either nesting of the two boxes earns full marks. Fatal errors (0 for the step): $\forall$I on a name that appears in an open assumption; an

∃E conclusion containing the witness.

#### 6.4.8.4 R. Review

**Exercise. R1**

Derive $\forall x\,(f(x) = x) \vdash \forall x\,(f(f(x)) = x)$.

*Solution.* The goal is a ∀: open a flagged box. The premise is instantiated twice, at the flag and at the compound term $f(x_0)$, the term the goal's instance contains.

| | | | |
|---|---|---|---|
| 1 | $\forall x\,(f(x) = x)$ | | |
| 2 | $x_0$ \| $f(x_0) = x_0$ | | ∀E 1 |
| 3 | $f(f(x_0)) = f(x_0)$ | | ∀E 1 |
| 4 | $f(f(x_0)) = x_0$ | | =E 2, 3 |
| 5 | $\forall x\,(f(f(x)) = x)$ | | ∀I 2–4 |

Line 3 instantiates at $f(x_0)$, which is free for $x$. Line 4 is transitivity, Proposition 5.7.1 (b), written as its one =E step: the equation of line 2 rewrites the last $f(x_0)$ of line 3. At line 5, $x_0$ is not in the premise or in line 5, and no assumption is open. The method is Section 5.3's "instantiate at the term you need", here a compound term, with the equality of Section 5.7.

**Exercise. R2**

Show that the list $P \leftrightarrow \neg Q,\ Q \leftrightarrow \neg R,\ R \leftrightarrow \neg P$ is inconsistent.

*Solution.* A list is inconsistent when $\bot$ is derivable from it (Definition 3.2.1). The derivation below uses no RAA: three ¬I boxes give $\neg R$, $\neg P$ and $\neg Q$, and each $\leftrightarrow$ E is used in whichever of its two forms (Definition 2.4.1) the line needs.

| | | |
|---|---|---|
| 1 | $P \leftrightarrow \neg Q$ | |
| 2 | $Q \leftrightarrow \neg R$ | |
| 3 | $R \leftrightarrow \neg P$ | |
| 4 | $\quad P$ | asm, for ¬I |
| 5 | $\quad \neg Q$ | ↔E 1, 4 |
| 6 | $\quad\quad R$ | asm, for ¬I |
| 7 | $\quad\quad \neg P$ | ↔E 3, 6 |
| 8 | $\quad\quad \bot$ | ¬E 4, 7 |
| 9 | $\quad \neg R$ | ¬I 6–8 |
| 10 | $\quad Q$ | ↔E 2, 9 |
| 11 | $\quad \bot$ | ¬E 10, 5 |
| 12 | $\neg P$ | ¬I 4–11 |
| 13 | $R$ | ↔E 3, 12 |
| 14 | $\quad Q$ | asm, for ¬I |
| 15 | $\quad \neg R$ | ↔E 2, 14 |
| 16 | $\quad \bot$ | ¬E 13, 15 |
| 17 | $\neg Q$ | ¬I 14–16 |
| 18 | $P$ | ↔E 1, 17 |
| 19 | $\bot$ | ¬E 18, 12 |

A TAUT line (Section 4.8) would do it in one step, since the three formulas have no row that makes them all true: an odd cycle of "is the negation of" is impossible.

**Exercise. R3**

You have derived $P \wedge (Q \vee R) \vdash (P \wedge Q) \vee (P \wedge R)$. Derive

$$\exists x\,(P(x) \wedge (Q(x) \vee R(x))) \vdash \exists x\,(P(x) \wedge Q(x)) \vee \exists x\,(P(x) \wedge R(x)),$$

and say which line is the propositional law.

*Solution.*

| | | | |
|---|---|---|---|
| 1 | | $\exists x\,(P(x) \wedge (Q(x) \vee R(x)))$ | |
| 2 | $x_0$ | $P(x_0) \wedge (Q(x_0) \vee R(x_0))$ | asm, for $\exists$E |
| 3 | | $(P(x_0) \wedge Q(x_0)) \vee (P(x_0) \wedge R(x_0))$ | TAUT 2 |
| 4 | | $P(x_0) \wedge Q(x_0)$ | asm, for $\vee$E |
| 5 | | $\exists x\,(P(x) \wedge Q(x))$ | $\exists$I 4 |
| 6 | | $\exists x\,(P(x) \wedge Q(x)) \vee \exists x\,(P(x) \wedge R(x))$ | $\vee$I 5 |
| 7 | | $P(x_0) \wedge R(x_0)$ | asm, for $\vee$E |
| 8 | | $\exists x\,(P(x) \wedge R(x))$ | $\exists$I 7 |
| 9 | | $\exists x\,(P(x) \wedge Q(x)) \vee \exists x\,(P(x) \wedge R(x))$ | $\vee$I 8 |
| 10 | | $\exists x\,(P(x) \wedge Q(x)) \vee \exists x\,(P(x) \wedge R(x))$ | $\vee$E 3, 4–6, 7–9 |
| 11 | | $\exists x\,(P(x) \wedge Q(x)) \vee \exists x\,(P(x) \wedge R(x))$ | $\exists$E 1, 2–10 |

Line 3, justified by TAUT on the skeleton $p \wedge (q \vee r) \models (p \wedge q) \vee (p \wedge r)$, is the propositional law Proposition 1.6.1 of Section 1.6; citing it as a derived rule (Definition 2.3.1) gives the same line. At line 11, $x_0$ is not in line 11 or in line 1, the premise, and no assumption is open. The quantifier steps frame the law: this is the propositional core of Section 6.3.